تمرین ۱ ۴۰۲۱۰۱۳۳۹ حامد باطانی

1)

a) $x_1[n] = (0.9)^n u[n] - (0.3)^n u[n]$

we know that for signal $x[n] = a^n u[n]$, z transform is: $X(z) = \sum_{n\in\mathbb{Z}} x[n] z^{-n} = \sum_{n=0}^{\infty} a^n z^{-n} = \frac{1}{1-az^{-1}} = \frac{z}{z-a}$

$$\rightarrow X_1(z) = \frac{z}{z-0.9} - \frac{z}{z-0.3} = \frac{0.6z}{(z-0.9)(z-0.3)} \quad |z|>|0.9|$$

2) from part a, $a^n u[n] \rightarrow \frac{z}{z-a}$

Property ① : ROC ↑

$x[n] \xrightarrow{z} X(z)$

$\Rightarrow n\,x[n] \xrightarrow{z} -z\frac{dX}{dz}$

$$\xrightarrow{\text{Property ①}} n a^n u[n] \xrightarrow{z} -z\frac{d}{dz}\left(\frac{z}{z-a}\right)$$

$$\rightarrow X_2(z) = \frac{az}{(z-a)^2} \quad , \quad |z|>|a| \text{ ROC}$$

Property ② :

$x[n] \xrightarrow{z} X(z)$

$\Rightarrow e^{jw_0 n} x[n] \xrightarrow{z} X(e^{-jw_0} z)$

3) $x[n] = a^n \cos w_0 n\, u[n]$

$\cos w_0 n = \left(e^{jw_0 n} + e^{-jw_0 n}\right)\frac{1}{2}$

$$\Rightarrow Z\{a^n \cos w_0 n\, u[n]\} = \frac{1}{2} Z\{e^{jw_0 n} a^n u[n]\} + \frac{1}{2} Z\{e^{-jw_0 n} a^n u[n]\}$$

$$\xrightarrow{\text{Property ②}} = \frac{e^{-jw_0} z}{e^{-jw_0} z - 1}\frac{1}{2} + \frac{1}{2}\frac{e^{jw_0} z}{e^{jw_0} z - 1} = \frac{1}{2}\left[\frac{z}{z - ae^{jw_0}} + \frac{z}{z - ae^{jw_0}}\right]$$

$$\rightarrow X(z) = \frac{z^2 - az\cos w_0 n}{z^2 - 2a\cos w_0 z + a^2} \quad , \quad |z|>|a| \text{ ROC}$$

h)

1) $G_1(z) = \frac{1}{z(z-0.7)} = \frac{A}{z} + \frac{B}{z-0.7}$

$A+B=0$, $-0.7A=1 \Rightarrow A = -\frac{10}{7}$, $B = \frac{10}{7}$

Property ③:

$x[n] \xrightarrow{z} X(z)$

$\Rightarrow x[n-n_0) \xrightarrow{z} z^{-n_0} X(z)$

$\delta[n] \xrightarrow{z} 1 \xRightarrow{\text{prop. 3}} \delta[n-n_0] \xrightarrow{z} z^{-n_0}$

$n_0 = 1 \Rightarrow \delta[n-1] = z^{-1}$

also we know that $a^n u[n] \xrightarrow{z} \frac{z}{z-a}$

$\xrightarrow{\text{prop 3}} a^{n-1} u[n-1] = z^{-1}\frac{z}{z-a} = \frac{1}{z-a} \to$

$z^{-1}\{G_1(z)\} = -\frac{10}{7}\delta[n-1] + \frac{10}{7}(0/7)^{n-1} u[n-1]$

2) $\frac{z^2+2z+1}{z^2-0.5z-0.12} = \frac{z^2+2z+1}{(z-0.75)(z+0.25)}$

$= 1 + \underbrace{\frac{2.5z+1.12}{(z-0.75)(z+0.25)}}_{G_a} = 1 + \frac{A}{(z-0.75)} + \frac{B}{(z+0.25)}$

$A = (z-0.75)\,G_a(z)\Big|_{z=0.75} \simeq 2/99$

$B = (z+0.25)\,G_a(z)\Big|_{z=0.25} \simeq 0.49$

$\to z^{-1}\{G_r\} = \delta[n] + 2.99(0.75)^{n-1} u[n-1] - 0.49(-0.25)^{n-1}$

$\times\, u[n-1]$

3) $G_3(z) = z^5$

$\delta[n] \xrightarrow{z} 1 \xrightarrow{\text{prop.③}} \delta[n+5] \xrightarrow{z} z^5$

$\to z^{-1}\{G_3\} = \delta[n+5]$

2. Stability of DT systems

a) $H(z) = \dfrac{1}{z^3 + 2.7z^2 + 2.8z + 1.5} = \dfrac{1}{P(z)}$

Bilinear Transformation : $s = c \cdot \dfrac{z-1}{z+1}$, $c = \dfrac{2}{T}$

the exact value of $c>0$, does not effect ~~[illegible]~~ the sign of real part, just scales it, so to simplify, $c=1$ $\to s = \dfrac{z-1}{z+1} \Rightarrow z = \dfrac{1+s}{1-s}$

$\to P(s) = \left(\dfrac{1+s}{1-s}\right)^3 + 2.7\left(\dfrac{1+s}{1-s}\right)^2 + 2.8\left(\dfrac{1+s}{1-s}\right) + 1.5$

$\xrightarrow[\text{of matlab}]{\text{with help}} = \dfrac{-0.4s^3 + 2s^2 - 1.6s + 8}{(1-s)^3} = \dfrac{Q(s)}{(1-s)^3}$

now we apply Routh-herwitz to $-Q(s)$

s^3	0.4	1.6
s^2	-2	-8
s^1	a_1	0
s^0	-8	

$a_1 = \dfrac{-2 \times 1.6 - (-8)(0.4)}{-2} = 0$

$\Rightarrow$ 1 sign change

$\to$ RHS root $\to$ unstable

b) $z^3 + 2.7z^2 + 2.8 + 1.5 = 0$

$\xrightarrow{\text{with matlab}}$ $z_1 \simeq -0.5$, $z_{2,3} = -1.1 \pm 0.6j$
"roots"

→ two poles outside $|z| = 1$ → unstable.

3 Discretization of CT controllers

a) as indicated in the slides $\omega_s = K\omega_b$, $35 \leq K < 70$
(note that we could just pick T = 1ms for both systems but that wouldn't seem like what TA wanted in this part)

ω_b:

$$G(s) = \frac{5s+17}{s+1} \rightarrow \tau = 1 \ , \ \omega_b = \frac{1}{\tau} = 1 \text{ rad/s}$$

$$H(s) = \frac{8(s+3)}{s^2+10s+29} \Rightarrow \begin{cases} \omega_n^2 = 29 \\ 2 \cdot \zeta\omega_n = 10 \end{cases} \rightarrow \zeta = 0.93 , \omega_n = 5.4$$

$\rightarrow \omega_b \simeq 4.02$ rad/s

we choose K = 70 → $\underline{\underline{T_G = \frac{2\pi}{70} \simeq 0.089}}$, $\underline{\underline{T_H = \frac{2\pi}{4.02} \simeq 0.02}}$

b) G: $p = -1$, $z = -3.4$. apply $z = e^{sT_G}$

$\rightarrow G(z) = \frac{1-0.74z^{-1}}{1-0.91z^{-1}} K$, DC match gain → $K = 5.5$

$H(s)$: $P_{1,2} = -5 \pm 2j$, $z = -3$ apply $z = e^{sT}$

$$\Rightarrow H(z) = \frac{(1 - 0.93z^{-1})z^{-1}}{1 - 1.8z^{-1} + 0.8z^{-2}} K, \text{ DC gain match} \rightarrow K \simeq 0.14$$

c) $c = \frac{2}{T} \rightarrow c_G = \frac{2}{T_G} \simeq 22.5$, $s = c\,\frac{z-1}{z+1}$

$$\xrightarrow{\text{apply}} G(z) \simeq \frac{5.6 - 4z^{-1}}{1 + (-0.91)z^{-1}}$$

H: $c_G = \frac{2}{T_H} = 100$, $s = c\,\frac{z-1}{z+1}$ $\xrightarrow{\text{apply}}$

$$H(z) \simeq \frac{0.08 + 0.004z^{-1} - 0.07z^{-2}}{1 - 1.8z^{-1} + 0.8z^{-2}}$$

d) $c = \frac{w_0}{\tan(w_0 T/2)} \rightarrow G$: $c_G = \frac{5}{\tan(\frac{5 \times 0.089}{2})}$

$\rightarrow c_G \simeq 22$, $s = c\,\frac{z-1}{z+1}$

$$\xrightarrow{\text{apply}} G(z) = \frac{5/5 - 3/98z^{-1}}{1 - 0.913z^{-1}}$$

H: $c_H = \frac{5}{\tan(\frac{5 \times 0.002}{2})} \simeq 99.9$

$$\rightarrow H(z) \simeq \frac{0.082 + 0.003z^{-1} - 0.08z^{-2}}{1 - 1.8z^{-1} + 0.8z^{-2}}$$